Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

# Eksamen


# Examination


## Module EBN111
**Elektrisiteit en Elektronika**

4 Junie 2010

## Module EBN111
**Electricity and Electronics**

4 June 2010

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

| **Eksamensinligting:**<br>***Exam information:*** | | | |
|---|---|---|---|
| Maksimum punte:<br>*Maximum marks:* | 90 | Volpunte:<br>*Full marks:* | 90 |
| Duur van vraestel:<br>*Duration of paper:* | 180 minute<br>*180 minutes* | Oopboek / toeboek:<br>*Open / closed book:* | Toe<br>*Closed* |
| Totale aantal bladsye (hierdie blad ingesluit):<br>*Total numer of pages (including this page):* | | 31 | |

**BELANGRIK- *IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   *The examination regulations of the University of Pretoria apply.*
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
   *All Questions must be answered on the ANSWER SHEET.*
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
   *Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
4. Polêre vorm: $A\angle\theta$ ; Reghoekige vorm: $a+jb$
   *Polar form:* $A\angle\theta$ *; Rectangular form:* $a+jb$

| | |
|---|---|
| Dosent(e):<br>*Lecturer(s):* | 1 J Joubert<br>2 D Bhatt<br>3 B Mabuza |

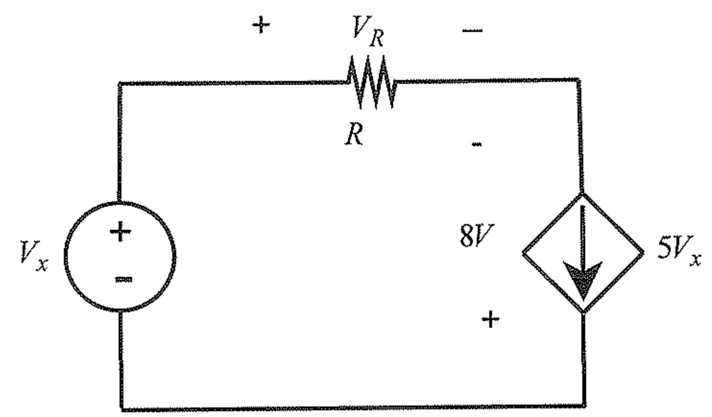
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding.
*Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.*

| | | | |
|---|---|---|---|
| Groephoof:<br>*Group Head:* | Prof J Joubert | Handtekening:<br>*Signature:* | |

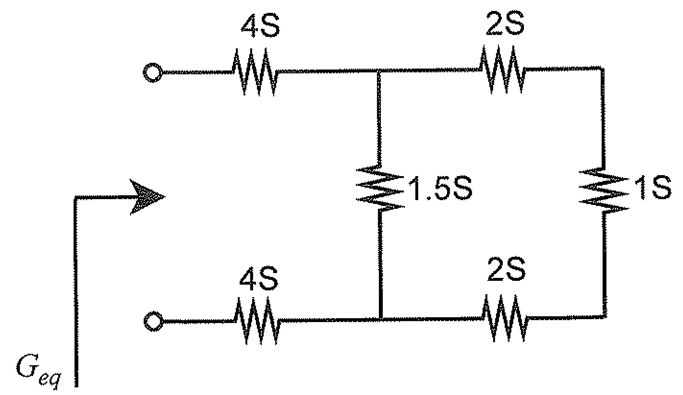
## VRAAG/QUESTION 1 [30]

| Vraag 1.1 | Question 1.1 |
|---|---|
| 'n Klein elektriese voertuig is toegerus met 'n 25 kW elektriese motor.<br>(a) Hoeveel energie (in MJ) word gebruik as die motor volspoed vir 3 ure lank loop. [1]<br>(b) As 'n enkele battery 15 kWh stoorkapasiteit het, hoeveel batterye word benodig om die kar vir 6 ure teen volspoed te laat loop. [1] | A small electric car is fitted with a 25 kW electric motor.<br>(a) How much energy (in MJ) is required for the motor to operate at full speed for 3 hours. [1]<br>(b) If a single battery has a store capacity of 15 kWh, how many batteries are needed for the motor to operate full speed for 6 hours. [1] |

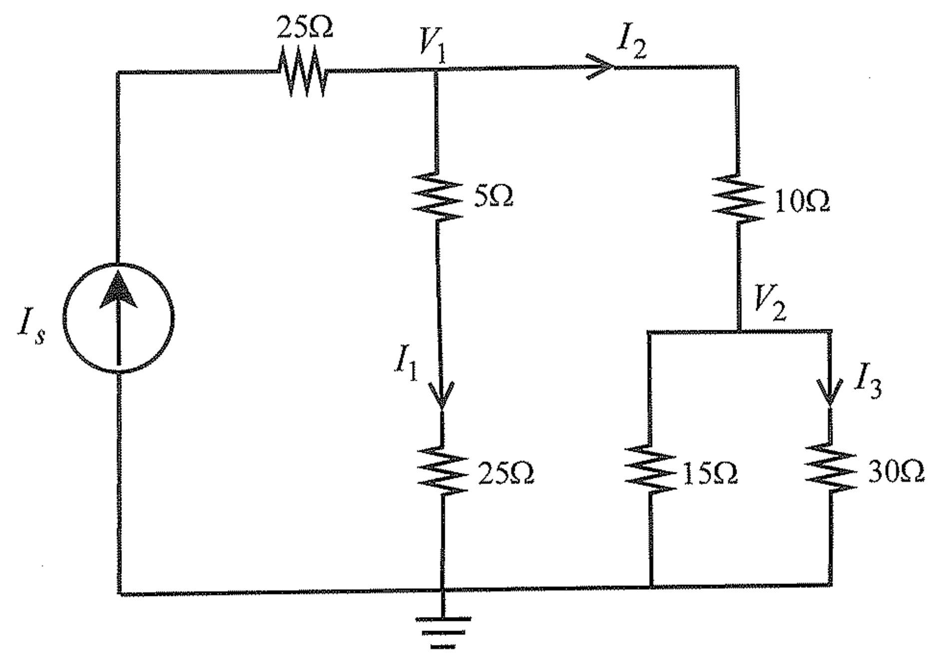
| Vraag 1.2 | Question 1.2 |
| --- | --- |
| (a) 'n Herlaaibare flitslig is in staat om 60 mA vir omtrent 10 ure lank te voorsien. Hoeveel lading word gelewer in daardie tyd. [1]<br>(b) Die lading wat by die positiewe terminaal van 'n element ingaan is $q = 100\sin(5t)$ mC, as die spanning oor die element $v = 2\cos(5t)$ V is. Bepaal die drywing geassosieer met die element by t = 0.2 s. [2] | (a) A rechargeable flashlight is capable of delivering 60 mA for about 10 hours. How much charge can it release during that time. [1]<br>(b) The charge entering the positive terminal of an element is given by $q = 100\sin(5t)$ mC, while the voltage across the element is $v = 2\cos(5t)$ V. Find the power associated with the element at t = 0.2 s. [2] |

| Vraag 1.3 | Question 1.3 |
|---|---|
|  | |
| Indien $V_x = 1$V;<br>(a) Bepaal die drywing ge-assosieer met die spanningsbron. [1]<br>(b) Bepaal die drywing ge-assosieer met die stroombron. [1] | If $V_x = 1$V;<br>(a) Determine the power associated with the voltage source. [1]<br>(b) Determine the power associated with the current source. [1] |

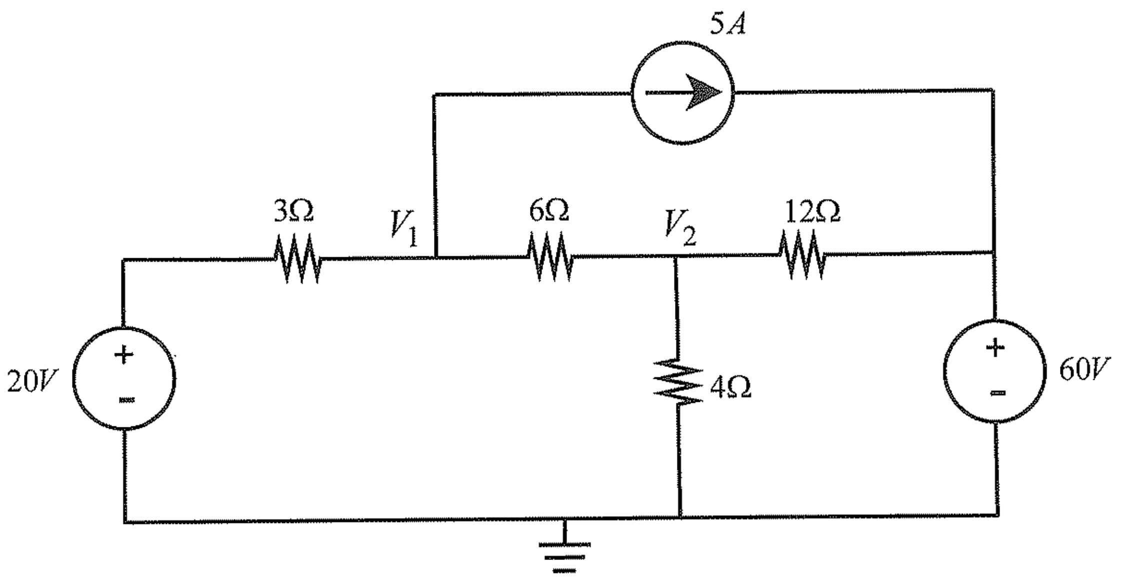
| Vraag 1.4 | Question 1.4 |
| --- | --- |




| | |
| --- | --- |
| Bepaal $G_{eq}$. [2] | Determine $G_{eq}$. [2] |

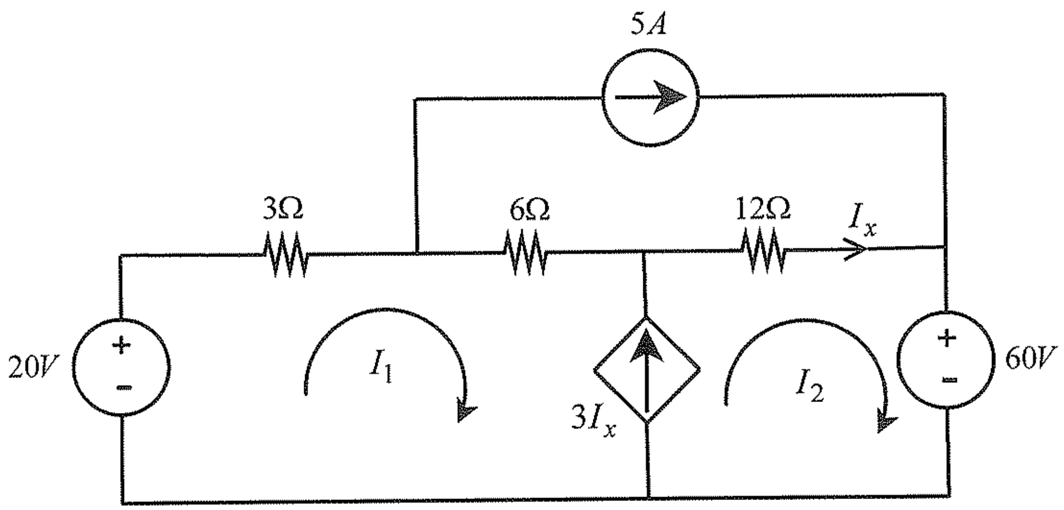
| Vraag 1.5 | Question 1.5 |
|---|---|
|  | |
| (a) Bepaal $I_3$ as $I_2 = 12$ mA. [1]<br>(b) Bepaal $V_2$ as $V_1 = 30$V. [1]<br>(c) Bepaal $I_3$ as $I_s = 60$ mA. [2] | (a) Determine $I_3$ if $I_2 = 12$ mA. [1]<br>(b) Determine $V_2$ if $V_1 = 30$V. [1]<br>(c) Determine $I_3$ if $I_s = 60$ mA. [2] |

| Vraag 1.6 | Question 1.6 |
|---|---|



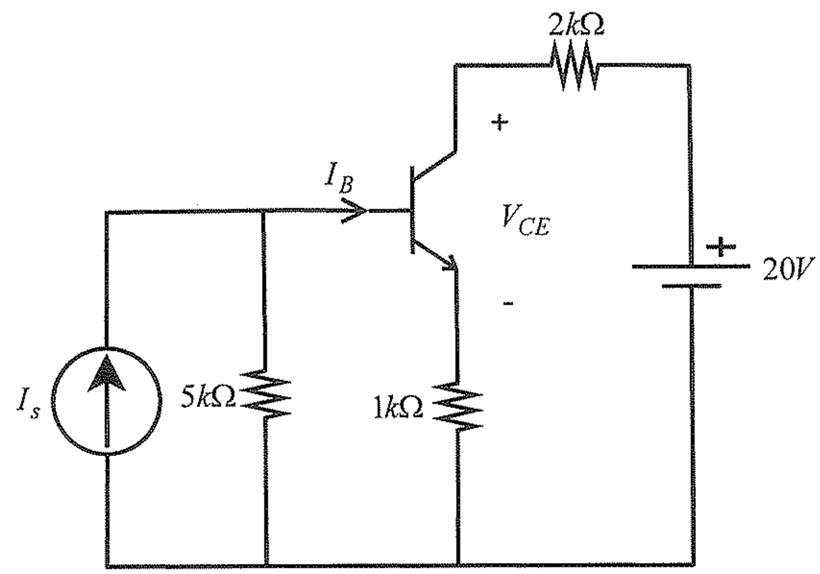

| Gebruik node-analise om twee vergelykings van die volgende vorm op te stel: $aV_1 - V_2 = b$ <br> $-2V_1 + cV_2 = d$ [4] | Use nodal analysis to formulate two equations of the following form: $aV_1 - V_2 = b$ <br> $-2V_1 + cV_2 = d$ [4] |
|---|---|

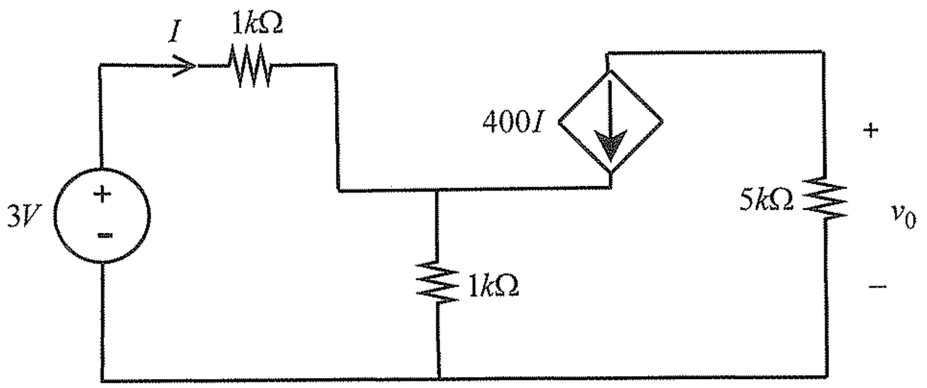
| Vraag 1.7 | Question 1.7 |
|---|---|



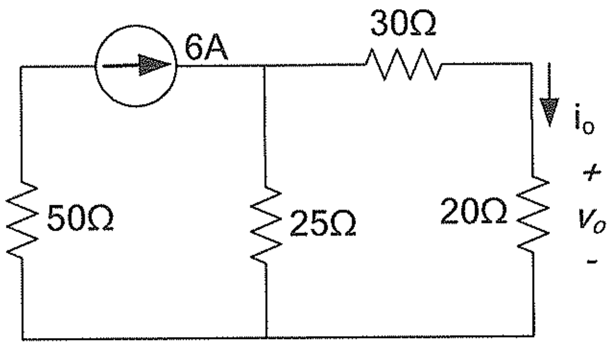

| | |
|---|---|
| Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel: $I_1 + eI_2 = f$ $9I_1 + gI_2 = h$ [4] | Use mesh analysis to formulate two equations of the following form: $I_1 + eI_2 = f$ $9I_1 + gI_2 = h$ [4] |

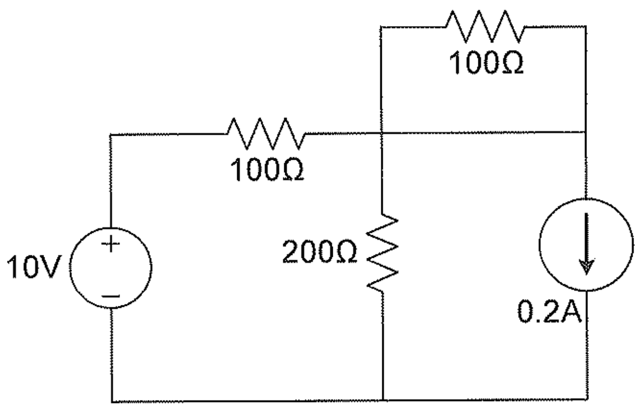
| Vraag 1.8 | Question 1.8 |
|---|---|
| Bepaal die determinant Δ en los die matriksvegelyking op om $v_2$ te bepaal. [2]<br>$$\begin{bmatrix} 2 & 0 & -2 \\ -1 & 3 & 0 \\ 2 & 1 & 0 \end{bmatrix} . \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 10 \end{bmatrix}$$ | Find the determinant Δ and solve the matrix equation to find $v_2$. [2]<br>$$\begin{bmatrix} 2 & 0 & -2 \\ -1 & 3 & 0 \\ 2 & 1 & 0 \end{bmatrix} . \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 10 \end{bmatrix}$$ |

<table>
<tr><th>Vraag 1.9</th><th>Question 1.9</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>(a) Bereken $I_B$ (in µA) as $I_s$ = 1mA, $\beta$ = 100 en $V_{BE}$ = 0.7V. [2]<br>(b) Bereken $V_{CE}$ as $I_B$ = 25 µA, $\beta$ = 100 en $V_{BE}$ = 0.7V. [2]</td><td>(a) Calculate $I_B$ (in µA) if $I_s$ = 1mA, $\beta$ = 100 and $V_{BE}$ = 0.7V. [2]<br>(b) Calculate $V_{CE}$ if $I_B$ = 25 µA, $\beta$ = 100 and $V_{BE}$ = 0.7V. [2]</td></tr>
</table>

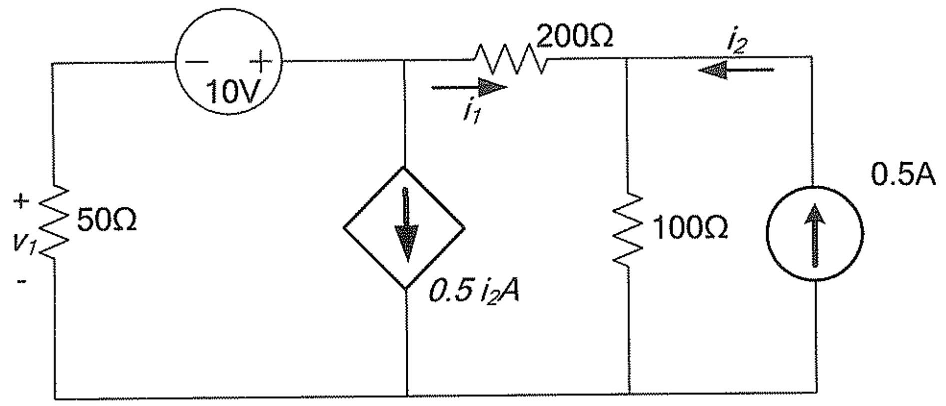
<table>
<tr><th>Vraag 1.10</th><th>Question 1.10</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bereken $I$ en $v_0$ in die vereenvoudigde transistor-baan. [3]</td><td>Calculate $I$ and $v_0$ in the simplified transistor circuit. [3]</td></tr>
</table>

# VRAAG/QUESTION 2 [25]

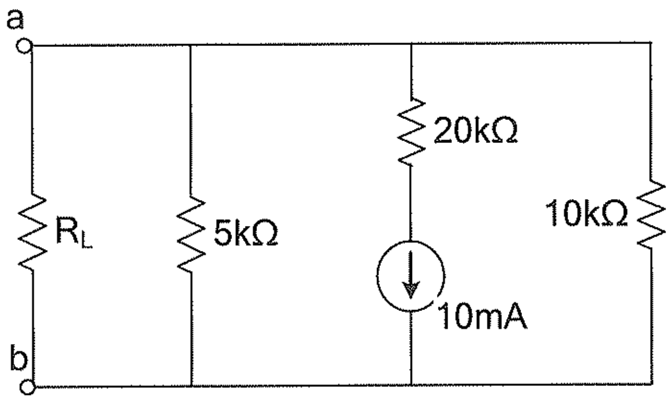
<table>
<tr><th>Vraag 2.1</th><th>Question 2.1</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Gebruik lineariteit om die spanning $v_o$ in die stroombaan te bereken (aanvaar aanvanklik $v_0 = 20$V). [2]</td><td>Use linearity to calculate the voltage $v_0$ in the circuit (initially assume $v_0 = 20$V). [2]</td></tr>
</table>

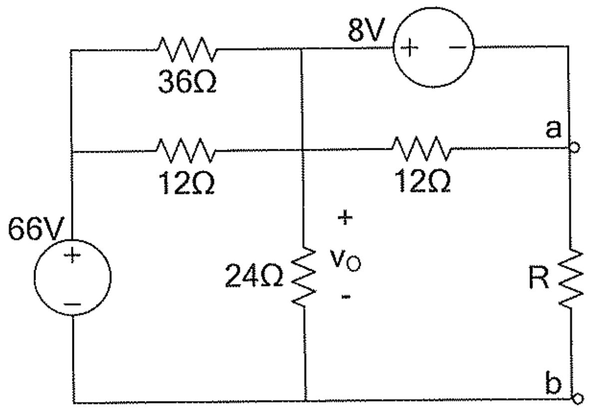
| Vraag 2.2 | Question 2.2 |
|---|---|
|  | |
| Vereenvoudig die stroombaan tot 'n enkele spanningsbron en 'n enkele weerstand. [2] | Reduce the circuit to a single voltage source and a single resistor. [2] |

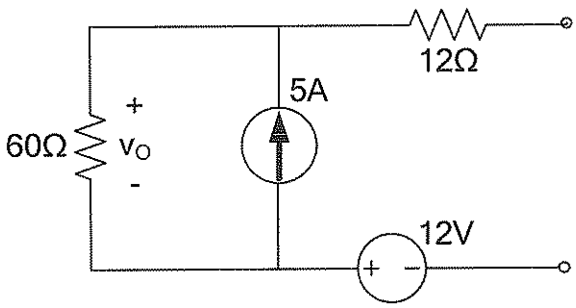
| Vraag 2.3 | Question 2.3 |
|---|---|

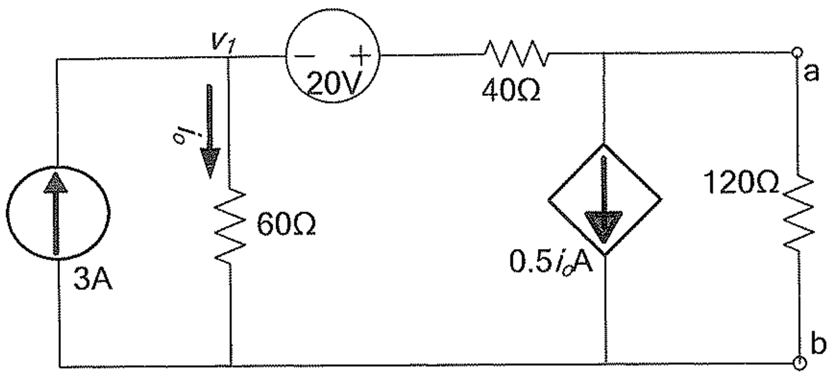


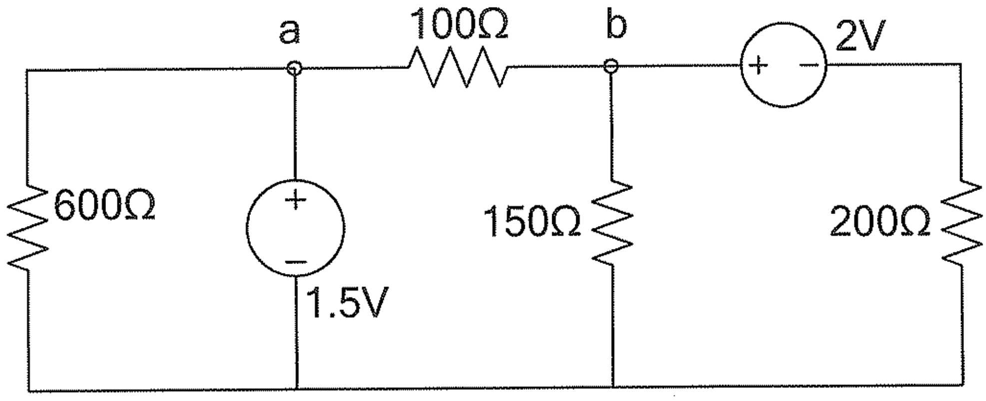

| | |
|---|---|
| Bereken:<br>(a) $v_1$ as gevolg van die 10V bron. [2]<br>(b) $i_1$ as gevolg van die 0.5A bron. [2] | Calculate:<br>(a) $v_1$ due to 10V source. [2]<br>(b) $i_1$ due to 0.5A source. [2] |

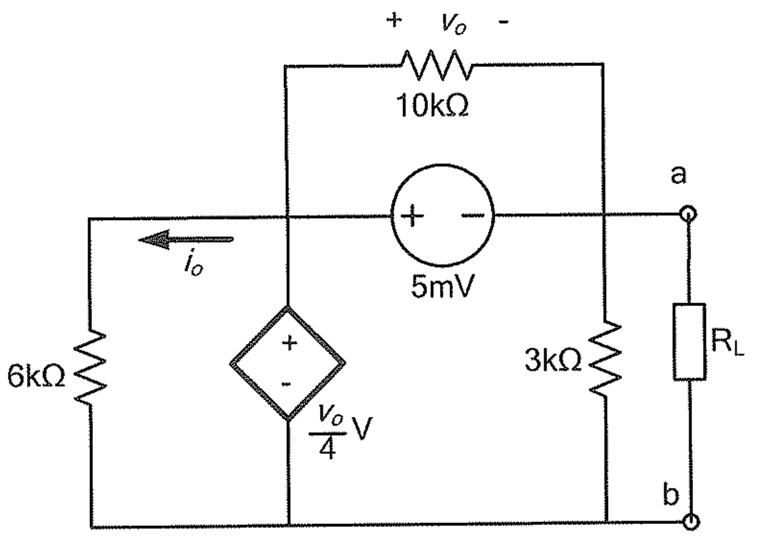
| Vraag 2.4 | Question 2.4 |
|---|---|
|  | |
| Wat is die optimum waarde van $R_L$ vir maksimum drywingsoordrag, en wat is die waarde van die maksimum drywing (in mW). [3] | What is the optimum value of $R_L$ for maximum power transfer, and what is the value of the maximum power (in mW). [3] |

<table>
<tr><th>Vraag 2.5</th><th>Question 2.5</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bereken die Thévenin ekwivalente weerstand $R_{TH}$ oor die las R (terminale a-b). [2]</td><td>Calculate the Thévenin equivalent resistance $R_{TH}$ across load R (terminal a-b). [2]</td></tr>
</table>

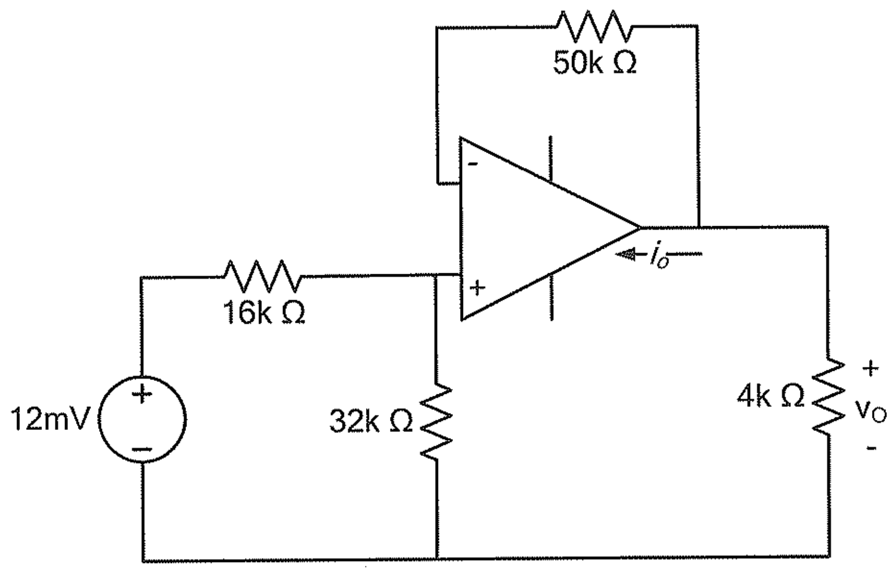
| Vraag 2.6 | Question 2.6 |
|---|---|
|  | |
| Bepaal die Thévenin ekwivalent. [4] | Determine the Thévenin equivalent. [4] |

| Vraag 2.7 | Question 2.7 |
|---|---|
|  | |
| Bepaal die Thévenin ekwivalent $R_{TH}$ ten opsigte van terminale a–b (verwyder die 120Ω). [2] | Find the Thévenin equivalent $R_{TH}$ with respect to terminals a–b (remove the 120Ω). [2] |

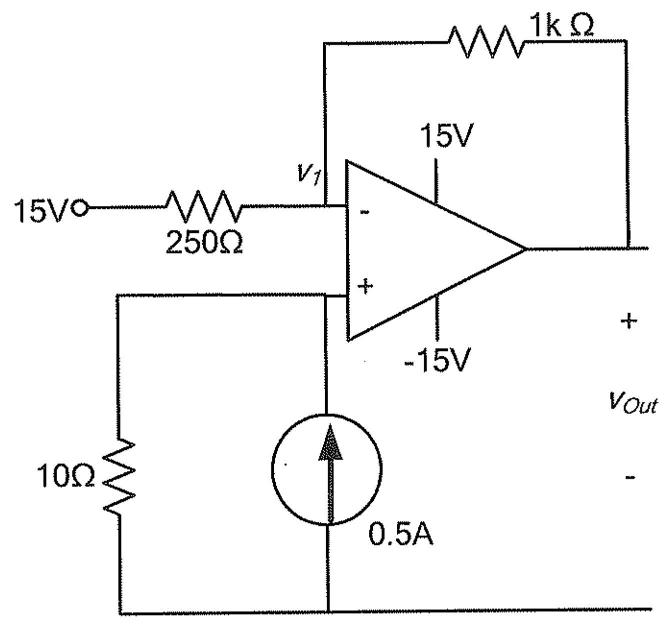
<table>
<tr><th>Vraag 2.8</th><th>Question 2.8</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bepaal die Norton ekwivalent $I_N$ and $R_N$ met verwysing tot terminale a–b (verwyder die 100Ω las). [4]</td><td>Find the Norton equivalent $I_N$ and $R_N$ with respect to terminals a–b (remove the 100Ω load). [4]</td></tr>
</table>

<table>
<tr><th>Vraag 2.9</th><th>Question 2.9</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bepaal die Norton ekwivalente weerstand $R_N$ met verwysing tot terminale a-b. [2]</td><td>Determine the Norton equivalent resistance $R_N$ with respect to terminals a-b. [2]</td></tr>
</table>

## VRAAG 3 – QUESTION 3 [15]

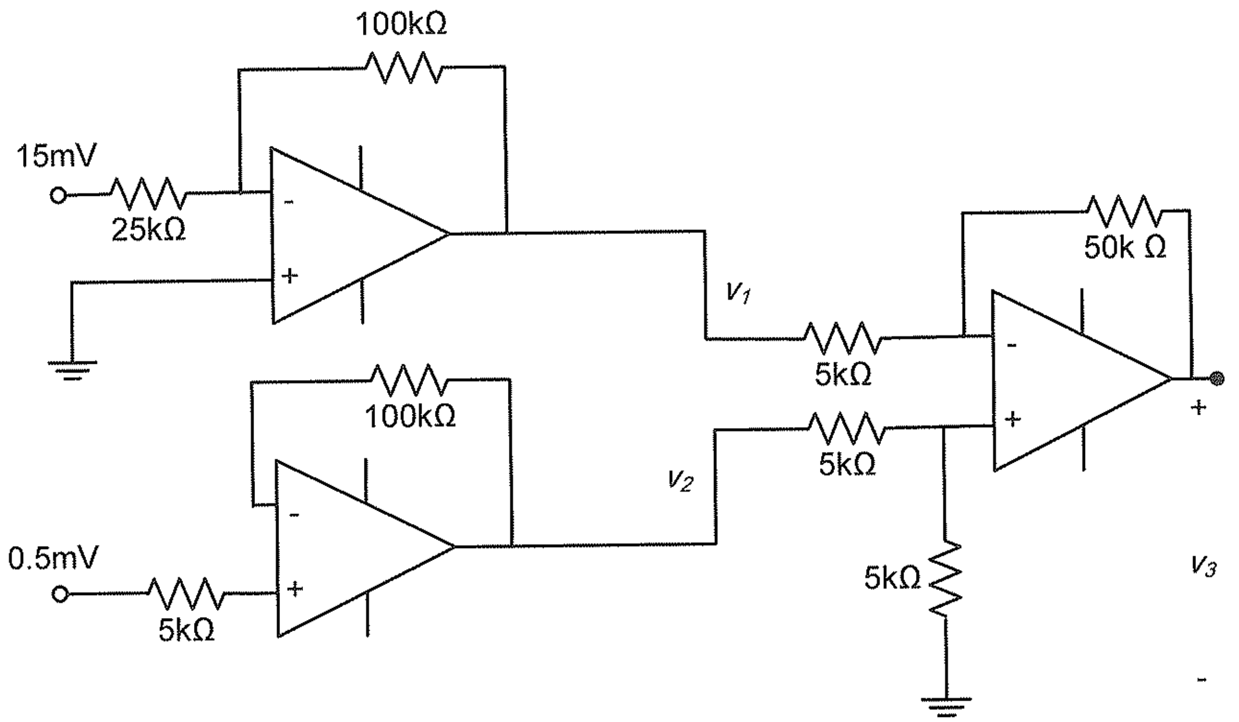
| Vraag 3.1 | Question 3.1 |
| --- | --- |
|  | |
| Bepaal die waarde van $v_O$ and $i_o$. [3] | Calculate the value of $v_O$ and $i_o$. [3] |

| Vraag 3.2 | Question 3.2 |
|---|---|

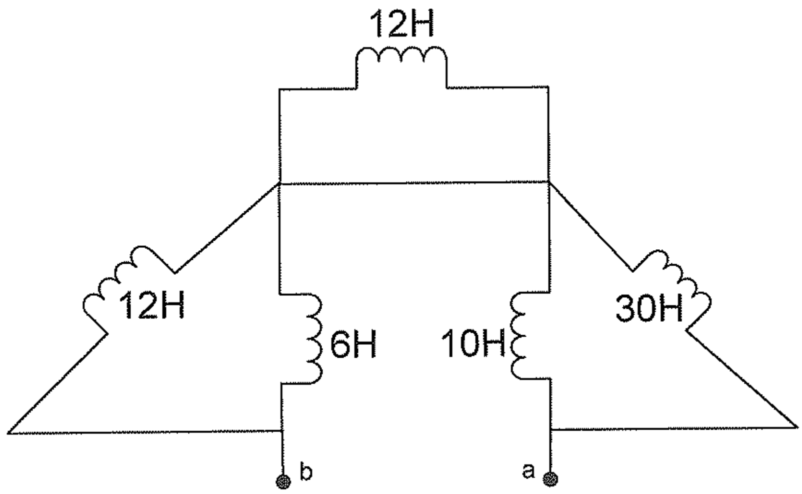


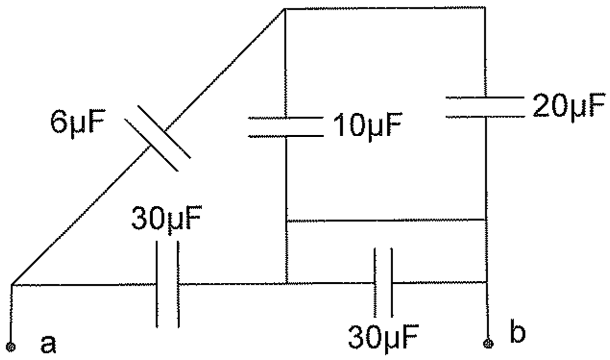

| Bepaal $V_{out}$. [2] | Determine $V_{out}$. [2] |
|---|---|

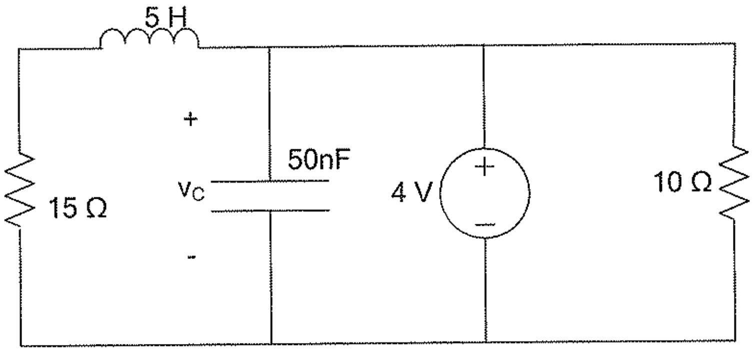
| **Vraag 3.3** | **Question 3.3** |
|---|---|




| Bepaal $V_1$, $V_2$, en $V_3$. [3] | Determine $V_1$, $V_2$, and $V_3$. [3] |
|---|---|

<table>
<tr><th>Vraag 3.4</th><th>Question 3.4</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bereken die ekwivalente induktansie tussen terminale a en b. [2]</td><td>Calculate the equivalent inductance between terminals a and b. [2]</td></tr>
</table>

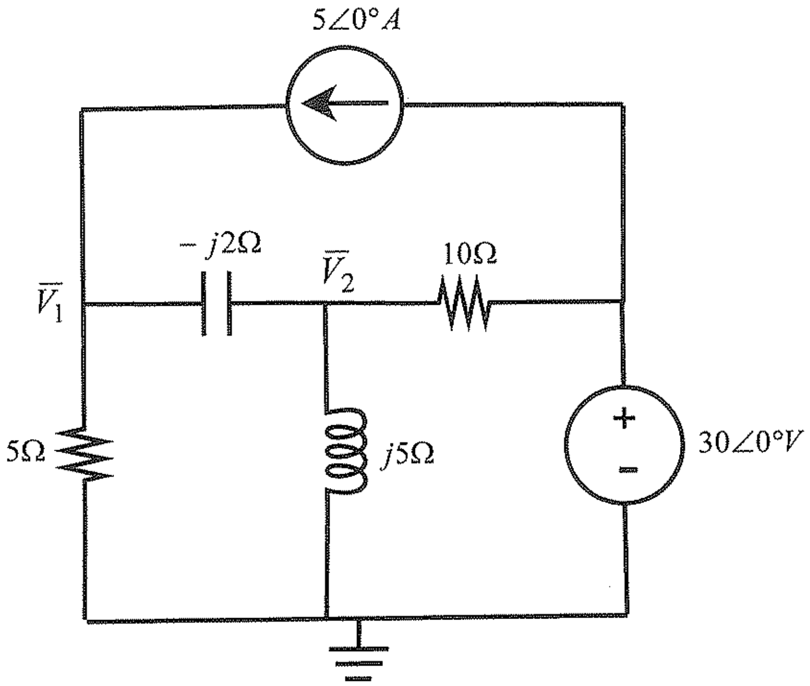
| Vraag 3.5 | Question 3.5 |
|---|---|
|   | |
| Bereken die ekwivalente kapasitansie tussen terminale $a$ en $b$. [2] | Calculate the equivalent capacitance between terminals $a$ and $b$. [2] |

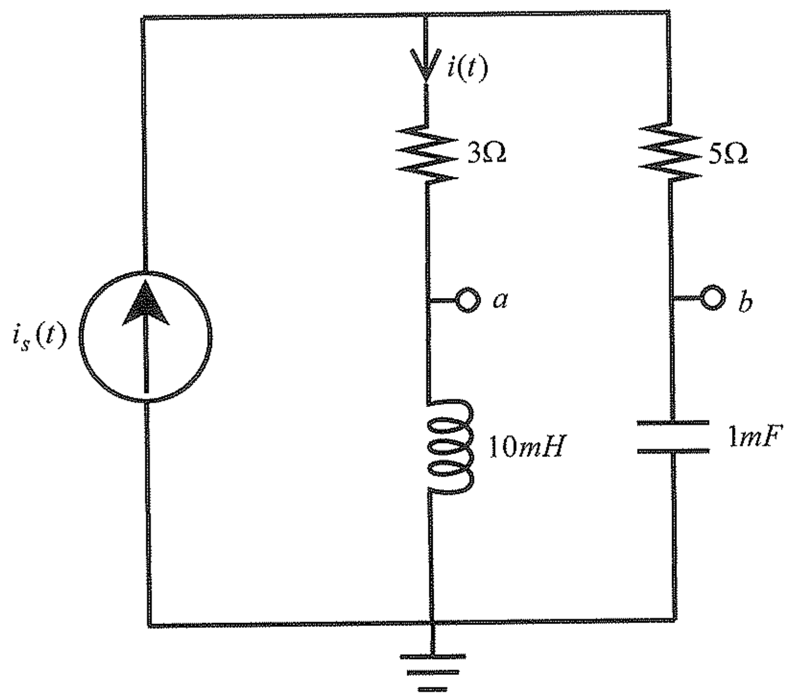
<table>
<tr><th>Vraag 3.6</th><th>Question 3.6</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Bepaal energiewaardes:<br>(a) gestoor in die kapasitor, [1]<br>(b) gestoor in die induktor, en [1]<br>(c) ge-assosieer met die 10Ω weerstand in een uur. [1]</td><td>Determine the energy:<br>(a) stored in the capacitor, [1]<br>(b) stored in the inductor, and [1]<br>(c) associated with the 10Ω resistor in one hour. [1]</td></tr>
</table>

# VRAAG 4 – QUESTION 4 [20]

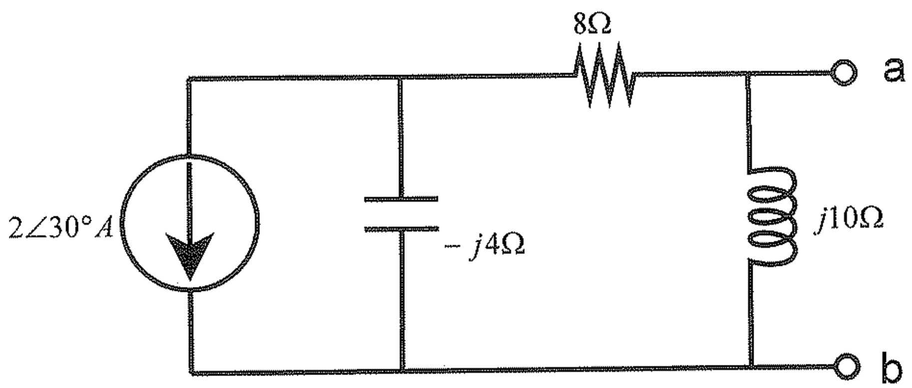
| Vraag 4.1 | Question 4.1 |
|---|---|
| (a) Skryf as 'n enkele cosinus uitdrukking: $v(t)=15\cos(\omega t+30°)-10\sin(\omega t-45°)$ V [2] | (a) Write as a single cosine expression: $v(t)=15\cos(\omega t+30°)-10\sin(\omega t-45°)$ V [2] |
| (b) Watter van die volgende fasors loop voor, en met hoeveel grade? $\overline{V}_1=10\angle-30°$, $\overline{V}_2=15\angle170°$ [2] | (b) Which of the following phasors is leading, and with how many degrees? $\overline{V}_1=10\angle-30°$, $\overline{V}_2=15\angle170°$ [2] |
| (c) Bepaal $\overline{V}_2$ (in reghoekige vorm): $\begin{bmatrix}1+j1 & -j\\ -j & 2-j2\end{bmatrix}\begin{bmatrix}\overline{V}_1\\ \overline{V}_2\end{bmatrix}=\begin{bmatrix}10\\ 0\end{bmatrix}$ [2] | (c) Determine $\overline{V}_2$ (in rectangular form): $\begin{bmatrix}1+j1 & -j\\ -j & 2-j2\end{bmatrix}\begin{bmatrix}\overline{V}_1\\ \overline{V}_2\end{bmatrix}=\begin{bmatrix}10\\ 0\end{bmatrix}$ [2] |

| Vraag 4.2 | Question 4.2 |
|---|---|
| (a) Evalueer die volgende (skryf antwoord in reghoekige vorm): $$\frac{\sqrt{(20+j10)}}{(3+j4)^*}$$ [2] | (a) Evaluate the following (write the answer in rectangular form): $$\frac{\sqrt{(20+j10)}}{(3+j4)^*}$$ [2] |
| (b) Evalueer die volgende (skryf antwoord in polêre vorm): $$\frac{(14-j9)}{(-2-j8)}+2e^{j30^\circ}$$ [2] | (b) Evaluate the following (write the answer in polar form): $$\frac{(14-j9)}{(-2-j8)}+2e^{j30^\circ}$$ [2] |

<table>
<tr><th>Vraag 4.3</th><th>Question 4.3</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Gebruik node-analise om twee gelyktydige vergelykings vir fasors $\overline{V}_1$ en $\overline{V}_2$ op te stel. Skryf in die vorm<br>$\overline{a}\overline{V}_1 + \overline{b}\,\overline{V}_2 = 50$<br>$\overline{c}\overline{V}_1 + \overline{d}\overline{V}_2 = 30$<br>met $\overline{a}$, $\overline{b}$, $\overline{c}$, and $\overline{d}$ komplekse koeffisiënte in reghoekige vorm. [4]</td><td>Use nodal analysis to set up two simultaneous equations for phasors $\overline{V}_1$ and $\overline{V}_2$. Write in the form<br>$\overline{a}\overline{V}_1 + \overline{b}\,\overline{V}_2 = 50$<br>$\overline{c}\overline{V}_1 + \overline{d}\overline{V}_2 = 30$<br>with $\overline{a}$, $\overline{b}$, $\overline{c}$, and $\overline{d}$ complex coefficients in rectangular form. [4]</td></tr>
</table>

<table>
<tr><th>Vraag 4.4</th><th>Question 4.4</th></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>Indien $i_s(t) = 3\cos(400t)$ A:<br>(a) Bereken i(t). [2]<br>(b) Bereken $Z_{ab}$ (in polêre vorm), die Thevenin impedansie t.o.v. terminale a-b. [2]</td><td>If $i_s(t) = 3\cos(400t)$ A:<br>(a) Determine i(t). [2]<br>(b) Determine $Z_{ab}$ (in polar form), the Thevenin impedance w.r.t. terminals a-b. [2]</td></tr>
</table>

| Vraag 4.5 | Question 4.5 |
| --- | --- |




| | |
| --- | --- |
| Bepaal $\overline{V}_{th}$ (die Thevenin spanning) t.o.v. terminale a-b. Skryf die antwoord in polêre vorm. [2] | Find $\overline{V}_{th}$ (the Thevinin voltage) w.r.t. terminals a-b. Write the answers in polar form. [2] |

EINDE - END